

CS570 Spring 2025: Analysis of Algorithms Exam I

	Points		Points
Problem 1	20	Problem 5	15
Problem 2	9	Problem 6	15
Problem 3	15	Problem 7	16
Problem 4	10		
Total 100			

First name	
Last Name	
Student ID	

Instructions:

1. This is a 2-hr exam. Closed book and notes. No electronic devices or internet access.
2. A single double sided 8.5in x 11in cheat sheet is allowed.
3. If a description to an algorithm or a proof is required, please limit your description or proof to within 150 words, preferably not exceeding the space allotted for that question.
4. No space other than the pages in the exam booklet will be scanned for grading.
5. If you require an additional page for a question, you can use the extra page provided within this booklet. However please indicate clearly that you are continuing the solution on the additional page.
6. Do not detach any sheets from the booklet.
7. If using a pencil to write the answers, make sure you apply enough pressure, so your answers are readable in the scanned copy of your exam.
8. Do not write your answers in cursive scripts.
9. This exam is printed double sided. Check and use the back of each page.

1) 20 pts (2 pts each)

Mark the following statements as **TRUE** or **FALSE** by circling the correct answer. No need to provide any justification.

[TRUE/FALSE]

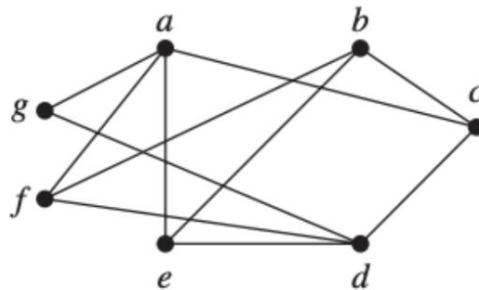
A Fibonacci Heap extract-min operation has a worst-case run time of $O(\log n)$.

[TRUE/FALSE]

Greedy algorithms make decisions to guarantee global optimization at each step.

[TRUE/FALSE]

The graph below is bipartite.



[TRUE/FALSE]

Dijkstra's algorithm will correctly find the shortest path in a graph containing negative edge weights, provided there are no negative cycles.

[TRUE/FALSE]

When performing DFS and BFS starting from a node u in a graph G , the number of levels in the DFS tree can never exceed the number of levels in the BFS tree.

[TRUE/FALSE]

If an edge is not part of any MST of G , then it must be the maximum weight edge on some cycle in G .

[TRUE/FALSE]

Suppose you have a directed graph G with a source vertex s , a target vertex t , and nonnegative edge weights. Furthermore, each edge has a distinct length that is a power of 2 (e.g., 1, 2, 4, 8, ..., and no two edges share the same power of 2). Then the shortest s - t path (in terms of total length) is guaranteed to be unique

[TRUE/FALSE]

Any function that is $\Omega(\log n)$ is also $\Omega(\log \log n)$.

[TRUE/FALSE]

We know that algorithm A has a worst-case running time of $O(n \log(n))$ and algorithm B has a worst-case running time of $O(n^2)$. It is possible for algorithm B to run faster than algorithm A on an input size n which is arbitrarily large.

[TRUE/FALSE]

When using the accounting method to find amortized costs for operations A, B, and C on data structure S, it is possible for all operations to end up with the same amortized cost even if their worst-case costs are different.

2) 9 pts **Select all correct answers!** 3 pts each. No partial credit

I. Which of the following algorithms use the divide and conquer approach?

- A) Heap Sort
- B) Merge Sort
- C) Prim's Algorithm
- D) Breadth-First Search (BFS)

II. Which of the following statements about binary min-heaps is incorrect?

- A) They are always complete binary trees
- B) Every parent node's key value is always less than its child nodes' key values
- C) They can be used to implement priority queues
- D) Merging two heaps of size n takes $O(\log n)$ worst-case time
- E) DECREASE-KEY takes $O(1)$ amortized time

III. Which of the following is true for asymptotic notations

- A) The worst-case time complexity of Merge sort is $\Omega(\log n)$
- B) If $f(n) = n \log(n^8)$ and $g(n) = \log n^n$ then $f(n) = O(g(n))$
- C) If $f(n) = o(g(n))$, then $g(n) = \Omega(f(n))$
- D) If $f(n) = O(n^2)$ and $g(n) = O(n^2)$, Then $f(n)/g(n) = O(1)$

3) 15 pts

Solve the following recurrence equations by giving tight theta-notation bounds in terms of n for sufficiently large n . Here, $T(\cdot)$ represents the running time of an algorithm. For each part below, either use the Master Method if it directly applies and clearly show which case applies and why, or explain why the Master Method does not apply and solve using the brute force method.

(a) $T(n) = 4T(n/2) + n^2 \log n$

(b) $T(n) = T(n/2) + T(n/4) + n^2$

(c) $T(n) = 16T(n/2) + 2^n$

4) 10 pts

Suppose you have a stable matching problem with n men and n women. You know all the preferences using which you run the G-S algorithm (with men proposing) to compute a stable matching. You also note down the number of times each man is rejected. Later, when you want to use the stable matching you computed, you realize that it is lost, and you have also lost some of the women's preference lists. Thus, all you are left with is

- 1) all of the men's preference lists
- 2) some but not all of the women's preference lists
- 3) the no. of rejections (r_i for each man i)

Devise an algorithm (with explanation and time complexity) to find a stable matching from the information available, or prove that it is not possible to compute one using the information available.

5) 15 pts

We have a list of N pets consisting of some dogs and others cats. We know the list has all dogs listed first, followed by all cats. You can make queries of the form “Is the n^{th} pet on the list a dog or a cat?” for any $n \leq N$. You want to determine how many dogs and how many cats are on the list. Devise an algorithm to do this by asking $\mathbf{o}(n)$ (little o) number of queries (along with time complexity analysis to show it has the desired efficiency). (Note that the simple algorithm making a pass over all pets would make $\mathbf{\theta}(n)$ queries, thus, not meeting the requirements).

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6) 15 pts

Consider an undirected, connected, weighted graph G (with both positive and negative edge costs).

a) True or False: Kruskal's algorithm may not correctly find an MST in the presence of negative cost edges. You need to explain your answer.

Now, for the graph G as above, we define a spanning network as a connected graph that contains a subset of the edges of G and spans across all nodes in G (but is not necessarily a tree, distinguishing it from a spanning tree). A Minimum Spanning Network (MSN) is a spanning network of minimum cost.

b) True or False: Every MST in G is an MSN. You need to explain your answer.

c) How would you modify Kruskal's algorithm to find an MSN? No proof is necessary.

7) 16 pts

You will be on vacation and need to make sure your dog will be taken care of at all times while you are away. You have a number of dog sitters that have provided you with their availability schedule, i.e. the intervals of time that they will be available to take care of your dog. Each of the n intervals i has a start time S_i and an end time E_i . To minimize your dog's stress level you want to plan a schedule for your dog sitters in such a way that you minimize the number of times your dog sees one dog sitter going away and another dog sitter replacing them. Design an algorithm to schedule your dog sitters in a way that there is always a dog sitter staying with your dog and minimizes the number of dog sitter exchanges. Your algorithm should run in $O(n \log n)$ in the worst case, and also be able to detect if such a schedule is not even feasible.

A) Describe your algorithm (6 pts)

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B) Prove the correctness of your algorithm (7 pts)

C) Analyze the complexity of your algorithm (3 pts)

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Additional Space

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